TOTALLY REAL GALOIS REALIZATIONS OF SL2 OVER F13 AND F19 BY PROJECTIVE STEMS AND CENTRAL LIFTING

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Allowed standard prerequisites (editorial).

- Automorphic tensor products, induction/base change and analytic Rankin-Selberg theory. Original complete bibliography and source-local auxiliary L-function arguments retained.

1. Introduction

1.1. **The inverse Galois problem for $\mathrm{SL}_2(\mathbb{F}_\ell)$.** The inverse Galois problem asks whether every finite group occurs as the Galois group over $\mathbb{Q}$. Zywina [Zyw15] proved that the projective groups $\mathrm{PSL}_2(\mathbb{F}_\ell)$ are Galois over $\mathbb{Q}$ for all primes $\ell \geqslant 5$. However, for the groups $\mathrm{SL}_2(\mathbb{F}_\ell)$, the answer is known only for $\ell = 5, 7, 11$. One reason the inverse Galois problem is harder for $\mathrm{SL}_2(\mathbb{F}_\ell)$ is parity. Two-dimensional representations attached to holomorphic modular eigenforms are odd, whereas a representation with image in $\mathrm{SL}_2(\mathbb{F}_\ell)$ has determinant one and is even.

A $\mathrm{PSL}_2(\mathbb{F}_\ell)$ extension $L/\mathbb{Q}$ gives a surjection $\varphi\colon \mathrm{Gal}_{\mathbb{Q}} \to \mathrm{PSL}_2(\mathbb{F}_\ell)$, and lifting it through the exact sequence

$$1 \longrightarrow \{\pm I\} \longrightarrow \mathrm{SL}_2(\mathbb{F}_\ell) \longrightarrow \mathrm{PSL}_2(\mathbb{F}_\ell) \longrightarrow 1 \tag{1.1}$$

is a *central embedding problem* whose obstruction lives in $H^2(\mathrm{Gal}_{\mathbb{Q}}, \{\pm 1\})$. That obstruction depends on the arithmetic of the particular field L and not on the abstract group $\mathrm{PSL}_2(\mathbb{F}_\ell)$. Klüners records $\mathrm{PSL}_2(\mathbb{F}_{11})$ extensions of $\mathbb{Q}$ that do not lift [Klü00] to $\mathrm{SL}_2(\mathbb{F}_{11})$. Hence, a realization of $\mathrm{SL}_2(\mathbb{F}_\ell)$ often needs a projective realization chosen for its local behavior.

The group $\mathrm{SL}_2(\mathbb{F}_5)$ was realized over $\mathbb{Q}$ by Sonn [Son80], and a regular realization was given by Mestre as a special case of his construction for $\widetilde{A}_n$ [Mes90]. Plans produced a generic extension [Pla07] for the same group. Mestre also constructed a regular realization of $\mathrm{SL}_2(\mathbb{F}_7)$ [Mes94b]. In [Klü00], Klüners found a degree-24 polynomial for $\mathrm{SL}_2(\mathbb{F}_{11})$, by a ray class field computation carried out under the generalized Riemann hypothesis, and obtained an unconditional regular realization by Mestre's method. We are not aware of a previously recorded realization for $\ell > 11$. Uttenthal observes that no explicit even irreducible mod ℓ (for $\ell \geqslant 13$) representation is available [Utt25, Remark 20]. Our main result is the following.

Theorem 1.1. *For $\ell = 13$ and for $\ell = 19$, there is a totally real Galois extension $M_\ell/\mathbb{Q}$ with $\mathrm{Gal}(M_\ell/\mathbb{Q}) \cong \mathrm{SL}_2(\mathbb{F}_\ell)$.*

1.2. **Projective stem fields.** Let ℓ be an odd prime, and let $\mathrm{PSL}_2(\mathbb{F}_\ell)$ act naturally on $\Omega_\ell = \mathbb{P}^1(\mathbb{F}_\ell)$. This action is faithful and transitive and has degree $\ell + 1$. If $L/\mathbb{Q}$ is Galois with group $\mathrm{PSL}_2(\mathbb{F}_\ell)$ and $B \leqslant \mathrm{PSL}_2(\mathbb{F}_\ell)$ is the stabilizer of a point of Ω_ℓ, we call the degree-$(\ell+1)$ field L^B a *projective stem field* of L.

Equivalently, let $f_\ell \in \mathbb{Q}[x]$ be irreducible of degree $\ell+1$, and let L_ℓ be its splitting field. Suppose that an isomorphism

$$\mathrm{Gal}(L_\ell/\mathbb{Q}) \simeq \mathrm{PSL}_2(\mathbb{F}_\ell)$$

identifies the action on the roots of f_ℓ with the natural action on $\mathbb{P}^1(\mathbb{F}_\ell)$. If α_ℓ is a root of f_ℓ, then

$$\mathbb{Q}(\alpha_\ell) = L_\ell^B,$$

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where B is the stabilizer of α_ℓ. Thus $\mathbb{Q}(\alpha_\ell)$ is a projective stem field. Klüners's degree 12 field for $\ell = 11$ is an example [Klü00]. Our arithmetic task is to construct such stem fields and verify the local conditions appearing in Böge's criterion.

Theorem 1.2. *Let $\ell \in \{13, 19\}$. For $\ell = 13$, let $f_\ell \in \mathbb{Z}[x]$ be the degree 14 polynomial displayed in Example 1.3, and for $\ell = 19$, let $f_\ell \in \mathbb{Z}[x]$ be the degree 20 polynomial in Appendix B. Then f_ℓ is irreducible, all $\ell + 1$ of its roots are real, and its splitting field L_ℓ satisfies*

$$\mathrm{Gal}(L_\ell/\mathbb{Q}) \cong \mathrm{PSL}_2(\mathbb{F}_\ell).$$

This isomorphism carries the action on the roots of f_ℓ to the action of $\mathrm{PSL}_2(\mathbb{F}_\ell)$ on $\mathbb{P}^1(\mathbb{F}_\ell)$. Moreover, for every odd rational prime q whose ramification index $e(q)$ in L_ℓ is even, the residue degree $f(q)$ satisfies

$$f(q) \text{ is odd} \qquad \Longleftrightarrow \qquad q \equiv 1 \pmod 4.$$

The last condition is exactly the condition (2) of Böge's criterion which we restate in Theorem 2.1. Assuming Theorem 1.2, we prove Theorem 1.1 in §2.5.

Example 1.3. For $\ell = 13$ one may take the following polynomial.

$$\begin{aligned} f_{13}(x) = {} & x^{14} - 6x^{13} - 204840441x^{12} - 1119125930364x^{11} \\ & + 4529702096707905x^{10} + 3815974962459546246x^9 \\ & - 18546069574715570089137x^8 - 45296853308950483736108589 6x^7 \\ & - 19266669911668478435133740133 3x^6 + 22095571675471627144021165180309 50x^5 \\ & + 14296168445224263309438957703242255 33x^4 \\ & - 38999201524351814558129637000444874470 36x^3 \\ & - 17691342639784521285621692548724503246355 97x^2 \\ & + 17925481734760709327105593729555826671414065 54x \\ & + 82582885667554637715998543349476798655946593258 9 \end{aligned} \tag{1.2}$$

By Proposition 3.1 and Lemma 3.2, its splitting field is a totally real $\mathrm{PSL}_2(\mathbb{F}_{13})$-extension of $\mathbb{Q}$, ramified exactly at

$$2, \quad 3, \quad 13, \quad 5119, \quad 1854089821344611, \quad 2086114950053051.$$

The degree-20 polynomial f_{19} has coefficients of up to 361 decimal digits, so we print it in Appendix B rather than here. By Lemma 4.8 its splitting field is ramified exactly at 17, at 5605357159, and at another prime of 100 digits.

Remark 1.4. The polynomials f_{13} and f_{19} have Galois groups $\mathrm{PSL}_2(\mathbb{F}_{13})$ and $\mathrm{PSL}_2(\mathbb{F}_{19})$. Böge's theorem only proves the existence of the fields M_ℓ of Theorem 1.1. We made the lift for $\ell = 13$ explicit using Crespo's method [Cre97], but the resulting polynomial has very large coefficients (on average order of 10^{180000}) and we do not include it in this paper. Accompanying files contain the polynomial with Galois group $\mathrm{SL}_2(\mathbb{F}_13)$.

1.3. **Constructing the polynomials.** We explain how the two polynomials f_ℓ were found. They were obtained through different methods.

For $\ell = 13$, we start from covers of $\mathbb{P}^1$, branched over four points, whose branch cycles lie in the conjugacy classes $2A, 2A, 2A, 6A$ of $\mathrm{PSL}_2(\mathbb{F}_{13})$. The associated absolute Hurwitz curve is a rational curve described by a degree 54 Belyi map. The corresponding inner Hurwitz curve is also rational and parametrizes a family of degree 14 covers with genus one source curves. Each rational point of the inner Hurwitz curve therefore determines one such cover defined over $\mathbb{Q}$. Specializing a chosen cover at a rational parameter for which all fourteen sheets are real produces f_{13}. This construction is not used in the proof of Theorem 1.2. The Galois group of f_{13} is certified independently in two ways. First, we certify it by a squarefree triple-sum resolvent of degree 364 that splits as $182 + 182$, and secondly, a Magma script confirms our proof. We describe the construction of f_{13} in Appendix A.

For $\ell = 19$, we begin with Yang's [Yan12] modular equation of level 19 on the Shimura curve $X_6^*(1)$. Its analytic branches are indexed by twenty level-19 Hecke cosets. An explicit reduction of the elliptic generators of the quaternion order proves that the monodromy of the geometric cover is $\mathrm{PGL}_2(\mathbb{F}_{19})$. We use the discriminant square class to cut out the $\mathrm{PSL}_2(\mathbb{F}_{19})$ subcover. Since Yang's cover has quite large coefficients, we first replace and identify it by an elliptic curve providing a degree 20 cover. After a quadratic base change that kills the sign character, it has harmonic branch values, i.e., with cross ratio -1. The quadratic base change gives a cover with a genus 2 domain. We move to another point on the rational inner Hurwitz curve it lies on, and get a specific rational cover F_*. A totally real specialization produces the polynomial f_{19}. Let G be the Galois group of f_{19}. We obtain an upper bound of G using the arithmetic monodromy of F_*. Factorization modulo small primes provide a lower bound, and we prove the equality $G = \mathrm{PSL}_2(\mathbb{F}_{19})$. An independent Magma calculation verifies the same conclusion.

Remark 1.5. Both of the families constructed above are branched over four points of $\mathbb{P}^1$, and in each case we exhibit a specialization for which the central embedding problem is solvable. This is the setting in which Klüners upgraded his $\ell = 11$ example to an unconditional *regular* realization, using Mestre's theorem on elements of $\mathrm{Br}_2(k(x))$ with four poles [Mes94a]. The similarity with Mestre's construction suggests that his argument may also apply to these two families, but we have not checked this.

1.4. **Overview.** In §2, we recall Böge's criterion, prove the necessary lemmas concerning the local invariants among others. Using them, we deduce Theorem 1.1 from Theorem 1.2. In §3, we prove Theorem 1.2 for $\ell = 13$. We first prove that the Galois group of the splitting field of f_{13} is $\mathrm{PSL}_2(\mathbb{F}_{13})$, which can also be checked using Magma. Afterwards we compute the local invariants of the field and invoke Böge's criterion. The Hurwitz construction behind f_{13} plays no part in that proof and is deferred to Appendix A. In §4, we prove Theorem 1.2 for $\ell = 19$. The construction of the polynomial defining the projective stem field comes from Yang's [Yan12] modular equations and we include the construction in the same section. In §5, we present a corollary using Ramakrishna's deformation theorem [Ram02]. In §6, we present the necessary scripts and talk about their reproducibility.

1.5. **Notation.** Throughout, $\ell \in \{13, 19\}$ and q denotes a rational prime. In the cases ℓ denotes an arbitrary prime, we will explicitly state so. We write $\mathrm{Gal}_{\mathbb{Q}} = \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$, $\Omega_\ell = \mathbb{P}^1(\mathbb{F}_\ell)$, and B for a point stabilizer in $\mathrm{PSL}_2(\mathbb{F}_\ell)$, that is, the image of a Borel subgroup. The fields

$$K_\ell \subset L_\ell \subset M_\ell$$

are, respectively, the projective stem field $\mathbb{Q}[x]/(f_\ell)$, its Galois closure, and the $\mathrm{SL}_2(\mathbb{F}_\ell)$ extension produced by Theorem 1.1.

For a rational prime q and a Galois extension $L/\mathbb{Q}$, all primes of L above q have the same ramification index and residue degree, and we denote them by $e(q)$ and $f(q)$. We reserve the subscripted symbols f_{13} and f_{19} for the two polynomials of Theorem 1.2. In a non-Galois field the factorization of q is recorded as a list of pairs (e_i, f_i), with multiplicities written as exponents.

Conjugacy classes of a finite group are named as in the Atlas [CCN+85], and a transitive group of degree n is named nTk as in the library of Hulpke [Hul23]. We write a cycle type multiplicatively, i.e., $1^2 2^6$ is a permutation of 14 points with two fixed points and six transpositions.

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https://github.com/eekarabiyik/SL2F13.

2. From a projective stem to $\mathrm{SL}_2(\mathbb{F}_\ell)$

2.1. **The embedding problem.** Let $L/\mathbb{Q}$ be Galois with group $\mathrm{PSL}_2(\mathbb{F}_\ell)$ and let $\varphi\colon \mathrm{Gal}_{\mathbb{Q}} \twoheadrightarrow \mathrm{PSL}_2(\mathbb{F}_\ell)$ be the associated surjection. A *solution* of the embedding problem attached to φ and the exact sequence (1.1) is a continuous lift of φ to $\mathrm{SL}_2(\mathbb{F}_\ell)$, equivalently a continuous homomorphism $\rho\colon \mathrm{Gal}_{\mathbb{Q}} \to \mathrm{SL}_2(\mathbb{F}_\ell)$ whose composition with the projection $\mathrm{SL}_2(\mathbb{F}_\ell) \to \mathrm{PSL}_2(\mathbb{F}_\ell)$ is φ. It is called *proper* if it is surjective, equivalently if the fixed field of $\ker(\rho)$ has Galois group $\mathrm{SL}_2(\mathbb{F}_\ell)$ over $\mathbb{Q}$.

The group $\mathrm{SL}_2(\mathbb{F}_\ell)$ is perfect so the extension (1.1) is nonsplit for every $\ell \geqslant 5$. The congruence $\ell \equiv 3, 5 \pmod 8$ that appears below controls a different kind of splitting. Assume $\ell \geqslant 11$. The natural action of $\mathrm{PSL}_2(\mathbb{F}_\ell)$ gives an embedding of $\mathrm{PSL}_2(\mathbb{F}_\ell)$ into $A_{\ell+1}$. Let $\widetilde{A}_{\ell+1}$ denote the unique nontrivial double cover of $A_{\ell+1}$, which is also called the *Schur cover*. Restricting this double cover to $\mathrm{PSL}_2(\mathbb{F}_\ell)$ produces a central extension with kernel $\{\pm 1\}$. When $\ell \equiv 3, 5 \pmod 8$, the resulting extension is $\mathrm{SL}_2(\mathbb{F}_\ell)$, and otherwise it is the split extension $\{\pm I\} \times \mathrm{PSL}_2(\mathbb{F}_\ell)$. In the first case, the obstruction to the embedding problem is computed by Serre's trace-form formula [Ser84]. Böge turns this obstruction into the arithmetic local criterion used below.

2.2. **Böge's criterion.**

Theorem 2.1 (Böge)**.** *Let ℓ be a prime with $\ell \equiv 3$ or $5 \pmod 8$ and let $L/\mathbb{Q}$ be a Galois extension with $\mathrm{Gal}(L/\mathbb{Q}) \cong \mathrm{PSL}_2(\mathbb{F}_\ell)$. The embedding problem attached to L and (1.1) has a proper solution if and only if*

(1) L is totally real, and

(2) for every odd prime q with $e(q)$ even,

$$f(q) \text{ is odd} \quad \Longleftrightarrow \quad q \equiv 1 \pmod 4.$$

This is an English restatement of Satz 1 of [Bög90, p. 153] which Klüners also states as [Klü00, Theorem 1.1]. There is no tameness hypothesis here. In particular, a wildly ramified prime is allowed, and one occurs for $\ell = 13$. There is also no condition at 2. Once the archimedean and the odd local invariants vanish, Brauer–Hasse–Noether reciprocity forces the dyadic invariant to vanish as well.

2.3. **Local invariants from a projective stem field.** Condition (2) of Theorem 2.1 is stated in terms of the ramification index and residue degree in the Galois extension L. However, direct computations in L are impractical, so instead we factor the prime in the degree $(\ell+1)$ projective stem field $K = L^B$. The following standard lemma translates this factorization into the orbit data of the inertia and decomposition groups, from which the required indices can be recovered. Compare Theorem 3.1 of [BK12]. We include a proof in the precise form needed here.

Lemma 2.2. *Let $L/\mathbb{Q}$ be a finite Galois extension whose group G acts faithfully and transitively on a finite set Ω, let B be a point stabilizer, and put $K = L^B$. Fix a prime of L above q and let $I \trianglelefteq D \leqslant G$ be its inertia and decomposition groups. Then the primes of K above q correspond to the D-orbits on Ω. An orbit $\mathcal{O} \ni \omega$ corresponds to a prime with*

$$e(\mathcal{O}) = |I\omega|, \qquad f(\mathcal{O}) = \frac{|D\omega|}{|I\omega|}. \tag{2.1}$$

Proof. Identify Ω with G/B. The primes of $K = L^B$ above q are indexed by the double cosets $D\backslash G/B$, hence by the D-orbits on G/B. If $\omega = gB$ then the stabilizers of ω in D and in I are $D \cap gBg^{-1}$ and $I \cap gBg^{-1}$, and local Galois theory gives

$$e(\mathcal{O}) = \frac{|I|}{|I \cap gBg^{-1}|} = |I\omega|, \qquad e(\mathcal{O})f(\mathcal{O}) = \frac{|D|}{|D \cap gBg^{-1}|} = |D\omega|.$$

This proves the assertion. Since I is normal in D, the I-orbits inside a fixed D-orbit all have the same size, so $e(\mathcal{O})$ does not depend on the choice of $\omega \in \mathcal{O}$. □

In our applications, we use Lemma 2.2 in the converse direction. Using GAP and PARI/GP we obtain the factorization type of q in K_ℓ and the different exponents. Afterwards, we do a finite search to list every pair $I \trianglelefteq D \leqslant \mathrm{PSL}_2(\mathbb{F}_\ell)$ producing that factorization type, and observe that in each case of

interest, there is only one possibility for $(e(q), f(q))$. For a tame prime, the search is easier since I and D/I are cyclic, and a Frobenius lift conjugates a generator of inertia to its qth power. For the only wild prime that occurs, at $q = 3$ in K_{13}, the search retains a superset of the arithmetically possible pairs but then we observe that still the same numerical pairs $(e(q), f(q))$ are obtained for every candidate pair in the superset.

Lemma 2.3. *With notation as in Lemma 2.2, a prime that is unramified in* K *is unramified in* L.

Proof. If q were ramified in L but not in K, then every I-orbit on Ω would be a singleton by (2.1), so I would act trivially on Ω. As G acts faithfully this forces $I = 1$.

□

2.4. **Totally real solutions.** The fixed field of a proper solution supplied by Böge's theorem does not have to be totally real. In our case, we can obtain one by a quadratic twist.

Lemma 2.4. *Let* $\ell > 3$ *and let* $L/\mathbb{Q}$ *be a totally real Galois extension with group* $\mathrm{PSL}_2(\mathbb{F}_\ell)$. *If the embedding problem attached to* L *and* (1.1) *has a proper solution, then it has a proper solution whose fixed field is totally real.*

Proof. Note that $\mathrm{SL}_2(\mathbb{F}_\ell)$ is perfect. Let $\varphi\colon \mathrm{Gal}_\mathbb{Q} \twoheadrightarrow \mathrm{PSL}_2(\mathbb{F}_\ell)$ be the surjection attached to the Galois extension L and let ρ be a proper solution to the embedding problem whose kernel cuts out a field with Galois group $\mathrm{SL}_2(\mathbb{F}_\ell)$. Let $c \in \mathrm{Gal}_\mathbb{Q}$ be a complex conjugation. Since L is a totally real field, $\varphi(c) = 1$ and $\rho(c) \in \{\pm I\}$. If $\rho(c) = I$, take χ to be the trivial character. Otherwise, let $\chi\colon \mathrm{Gal}_\mathbb{Q} \to \{\pm I\}$ be the quadratic character of an imaginary quadratic field. In both cases put $\rho^\chi(g) = \chi(g)\rho(g)$. Since χ takes values in the center, the projection of ρ^χ is still φ and ρ^χ is again a solution to the embedding problem. We have $\rho^\chi(c) = I$ by construction.

It remains to show that ρ^χ is still surjective. Its image maps onto $\mathrm{PSL}_2(\mathbb{F}_\ell)$, so it is either all of $\mathrm{SL}_2(\mathbb{F}_\ell)$ or a complement of $\{\pm I\}$, in which case its index is 2. The latter is impossible because a perfect group has no subgroup of index 2. Since all complex conjugations in $\mathrm{Gal}_\mathbb{Q}$ are conjugate, the fixed field of $\ker \rho^\chi$ is totally real. □

2.5. **Proof of Theorem 1.1.** Let $\ell \in \{13, 19\}$ and let L_ℓ be the splitting field of f_ℓ. By Theorem 1.2, we have $\mathrm{Gal}(L_\ell/\mathbb{Q}) \cong \mathrm{PSL}_2(\mathbb{F}_\ell)$, and L_ℓ is totally real. The last assertion of Theorem 1.2 satisfies condition (2) of Böge's criterion 2.1 and since $13 \equiv 5 \pmod 8$ and $19 \equiv 3 \pmod 8$, Theorem 2.1 applies and yields a proper solution ρ of the embedding problem attached to L_ℓ. The fixed field M_ℓ of $\ker \rho$ has $\mathrm{Gal}(M_\ell/\mathbb{Q}) \cong \mathrm{SL}_2(\mathbb{F}_\ell)$, and it can be chosen totally real by Lemma 2.4.

□

The next two sections prove Theorem 1.2.

3. The case $\ell = 13$

3.1. **The Galois group of f_{13}.** The argument in this subsection uses nothing but the coefficients of f_{13}. Proposition 3.1 proves the needed result using group theory, but it can be explicitly checked using Magma, cf. §6.

Proposition 3.1. *The polynomial* f_{13} *written in Example 1.3 is irreducible, all fourteen of its roots are real, and its splitting field* L_{13} *satisfies*

$$\mathrm{Gal}(L_{13}/\mathbb{Q}) \cong \mathrm{PSL}_2(\mathbb{F}_{13})$$

in the natural action of $\mathrm{PSL}_2(\mathbb{F}_{13})$ *on* $\mathbb{P}^1(\mathbb{F}_{13})$.

Proof. Modulo 5 the reduction of f_{13} is a squarefree product of two irreducible factors of degree 7. Modulo 23 it factors as

$$(x+8)(x^{13} + 9x^{12} + 17x^{11} + 19x^{10} + 5x^9 + 15x^8 + 6x^7 + 16x^6 + 19x^5 + 15x^4 + 12x^3 + 4x^2 + 21x + 7)$$

with the degree 13 factor irreducible. A nontrivial factor of f_{13} over $\mathbb{Q}$ would have degree 7 by the first reduction and degree 1 or 13 by the second, so f_{13} is irreducible and its Galois group G is transitive of degree 14.

The discriminant of f_{13} is a nonzero square, so $G \leqslant A_{14}$. The reduction modulo 23 is squarefree, so $23 \nmid \operatorname{disc}(f_{13})$ and so G contains an element of cycle type $1 \cdot 13$, that is, a 13-cycle. Among the 63 transitive groups of degree 14 classified by Butler [But93] exactly two are contained in A_{14} and contain a 13-cycle, namely

$$14T30 \cong \mathrm{PSL}_2(\mathbb{F}_{13}) \qquad \text{and} \qquad 14T62 = A_{14}.$$

We verified this by inspecting the entire library [Hul23] in GAP.

Let $r_1, \ldots, r_{14}$ be the roots of f_{13}. To each three-element set $\{i, j, k\}$ attach the sum $r_i + r_j + r_k$, and put all these sums into one polynomial

$$R_3(y) = \prod_{1 \leqslant i < j < k \leqslant 14} \bigl(y - (r_i + r_j + r_k)\bigr),$$

which is the triple sum resolvent. We compute it exactly from the coefficients of f_{13} with no numerical approximation of any of the roots.

Let $s_m = \sum_i r_i^m$. For $a \in \{1, 2, 3\}$, define

$$P_a(z) = \sum_{m=0}^{364} \frac{a^m s_m}{m!} z^m.$$

Then, modulo z^{365},

$$\sum_{i<j<k} \exp\bigl((r_i + r_j + r_k)z\bigr) = \frac{P_1(z)^3 - 3P_1(z)P_2(z) + 2P_3(z)}{6}.$$

All equalities here are taken modulo z^{365}. Newton identities compute $s_1, \ldots, s_{364}$ directly from f_{13}. For the sum, $m!$ times the coefficient of z^m is

$$\sum_{i<j<k} (r_i + r_j + r_k)^m,$$

the m-th power sum of the 364 roots of R_3. Applying Newton identities a second time therefore reconstructs every coefficient of R_3.

Each coefficient of R_3 is a symmetric polynomial with integer coefficients in $r_1, \ldots, r_{14}$. The fundamental theorem of symmetric polynomials expresses it as an integer polynomial in the elementary symmetric functions of the r_i and consequently $R_3 \in \mathbb{Z}[y]$.

We show, using Sage, that R_3 is squarefree and is the product of two irreducible polynomials over $\mathbb{Q}$, each of degree 182. Thus

$$\{i, j, k\} \longmapsto r_i + r_j + r_k$$

is a G-equivariant bijection from the three-subsets of the roots to the roots of R_3. Under this bijection, the irreducible factors of R_3 correspond exactly to the G-orbits. Hence G has two orbits, both of size 182, on the three-subsets.

The group A_{14} is transitive on all 364 three-subsets. In contrast, the natural degree-14 action of $\mathrm{PSL}_2(\mathbb{F}_{13})$ has precisely two orbits of size 182. As a result, we conclude that $G \cong \mathrm{PSL}_2(\mathbb{F}_{13})$ in its natural action. Sturm arithmetic in the accompanying Sage file gives fourteen real roots [BPR06, §2.2.2].

Additionally, an exact Magma script in the level 13 code package proves irreducibility and, using `GaloisProof`, identifies the Galois group in its degree-14 action with $\mathrm{PSL}_2(\mathbb{F}_{13})$. □

3.2. **Ramification.** Write $K_{13} = \mathbb{Q}[x]/(f_{13})$ and set

$$p_1 = 1854089821344611, \qquad p_2 = 2086114950053051,$$

both prime. We have:

$$\operatorname{disc}(K_{13}) = 2^{22}\, 3^{22}\, 13^6\, 5119^6\, p_1^6\, p_2^6. \tag{3.1}$$

Lemma 3.2. *The primes ramified in* L_{13} *are exactly* $2, 3, 13, 5119, p_1, p_2$*, and at each odd one the pair* $(e(q), f(q))$ *is as in Table 1.*

q	type of q in K_{13}	(I, D)	$(e(q), f(q))$
2	$(1,2), (6,1)^2$	—	not needed
3	$(2,1), (3,2), (6,1)$	(S_3, D_{12})	$(6,2)$
13	$(1,1)^2, (2,3)^2$	(C_2, C_6)	$(2,3)$
5119, p_1, p_2	$(1,2), (2,1)^2, (2,2)^2$	$(C_2, C_2 \times C_2)$	$(2,2)$

TABLE 1. Local data for K_{13} and for its Galois closure L_{13}.

Proof. We construct the maximal order of K_{13} using PARI/GP, certify it with `nfcertify`, and also confirm the discriminant (3.1). Factoring the six ramified primes in this maximal order gives the second column of Table 1.

We now apply Lemma 2.2 with $G = \mathrm{PSL}_2(\mathbb{F}_{13})$ acting on fourteen points, and enumerate the pairs $I \trianglelefteq D \leqslant G$ which can realize each factorization type. At $q = 3$ the enumeration retains a superset of the possible wild pairs. For each odd prime q in Table 1, these enumerated sets contain only one conjugacy class of pairs (I, D). In every case, we obtain the invariants appearing in Table 1.

Every prime ramified in K_{13} ramifies in L_{13}, and by Lemma 2.3 there is no further ramification, proving the assertion. □

Only the odd primes with $e(q)$ even are considered for Theorem 2.1, and those are $3, 13, 5119, p_1$ and p_2. For them,

q	3	13	5119	p_1	p_2
$f(q)$	2	3	2	2	2
$q \bmod 4$	3	1	3	3	3

so $f(q)$ is odd exactly when $q \equiv 1 \pmod 4$. Lemma 3.2 and Proposition 3.1 together prove Theorem 1.2 for $\ell = 13$.

4. The case $\ell = 19$

For $\ell = 19$, we include the construction of the polynomial here, as it involves some Shimura curve results. We start with Yang's [Yan13] cover, which comes from Hecke correspondences, with geometric monodromy $\mathrm{PGL}_2(\mathbb{F}_{19})$. After a quadratic base change one obtains a cover with monodromy $\mathrm{PSL}_2(\mathbb{F}_{19})$. We transform this pullback cover into a form that allows better computations and obtain the polynomial f_{19} by a precise search.

4.1. Yang's Hecke cover.

Let $\mathcal{B} = (-1, 3)_{\mathbb{Q}}$ be the indefinite quaternion algebra of discriminant 6, so that

$$I^2 = -1, \qquad J^2 = 3, \qquad IJ = -JI,$$

and let

$$\mathcal{O} = \mathbb{Z} + \mathbb{Z}I + \mathbb{Z}J + \mathbb{Z}\tfrac{1+I+J+IJ}{2}$$

be a maximal order. The full Atkin–Lehner quotient $X_6^*(1)$ of the Shimura curve $X_6(1)$ has genus 0. Yang [Yan13] fixes a Hauptmodul T on $X_6^*(1)$ by requiring the elliptic points of orders 6, 2 and 4 to lie at 0, -540 and ∞. We use Yang's modular equation algorithm [Yan12] to produce a level 19 modular equation associated with this Hauptmodul. Regarding it as a polynomial in z over $\mathbb{Q}(T)$ and dividing by its leading coefficient, we obtain a monic polynomial $P_0(T, z) \in \mathbb{Q}(T)[z]$ of degree 20 in z. More precisely, the level-19 Hecke double coset is the disjoint union of twenty right cosets. Let $\mathcal{C}_{19}$ denote the set of

these cosets. On a simply connected open set of the T-line, each $c \in \mathcal{C}_{19}$ determines an analytic branch z_c. We have

$$P_0(T, z) = \prod_{c \in \mathcal{C}_{19}} (z - z_c) \in \mathbb{Q}(T)[z]. \tag{4.1}$$

Thus, for a generic point $x \in X_6^*(1)$, the roots of $P_0(T(x), z)$ are the Hauptmodul values of the twenty points related to x by the level-19 Hecke correspondence. Clearing denominators and removing the common content gives the primitive integral model $P(T, z) \in \mathbb{Z}[T, z]$ of bidegree $(6, 20)$. We use this integral model P for the rest of the paper. Its exact coefficients can be found in the accompanying files. The following lemma helps us identify the monodromy of P.

Lemma 4.1. *Choose a splitting $\mathcal{O} \otimes \mathbb{Z}_{19} \simeq M_2(\mathbb{Z}_{19})$. It gives natural equivariant bijections*

$$\mathcal{C}_{19} \;\simeq\; \mathrm{PGL}_2(\mathbb{F}_{19})/B(\mathbb{F}_{19}) \;\simeq\; \mathbb{P}^1(\mathbb{F}_{19}).$$

Under these bijections, analytic continuation along a loop around an elliptic point permutes the roots z_c through the image in $\mathrm{PGL}_2(\mathbb{F}_{19})$ of a generator of the corresponding elliptic stabilizer.

Proof. Put $\mathcal{O}_{19} = \mathcal{O} \otimes \mathbb{Z}_{19}$, and let $\mathcal{E}_{19} \subset \mathcal{O}_{19}$ be the local Eichler order of level 19. Define

$$K = \mathcal{O}_{19}^\times / \mathbb{Z}_{19}^\times, \qquad K_0(19) = \mathcal{E}_{19}^\times / \mathbb{Z}_{19}^\times.$$

Reduction modulo 19 maps K onto $\mathrm{PGL}_2(\mathbb{F}_{19})$, and $K_0(19)$ is the inverse image of the upper triangular Borel subgroup $B(\mathbb{F}_{19})$. The kernel of reduction is contained in $K_0(19)$, so reduction induces a bijection

$$K/K_0(19) \;\simeq\; \mathrm{PGL}_2(\mathbb{F}_{19})/B(\mathbb{F}_{19}).$$

The cosets $\mathcal{C}_{19}$ indexing Yang's double-coset product are right cosets, and hence form the set $K_0(19)\backslash K$. We identify this set with the left-coset space above by inversion:

$$K_0(19)g \longmapsto g^{-1}K_0(19).$$

Under this identification, right multiplication by k corresponds to left multiplication by k^{-1}. We use this left-coset convention below. Let L_0 be the line spanned by $(1, 0)$ in $\mathbb{F}_{19}^2$. Its stabilizer in $\mathrm{PGL}_2(\mathbb{F}_{19})$ is $B(\mathbb{F}_{19})$, so

$$gB(\mathbb{F}_{19}) \longmapsto gL_0$$

is a well-defined equivariant bijection from the displayed coset space to the set of lines in $\mathbb{F}_{19}^2$, namely $\mathbb{P}^1(\mathbb{F}_{19})$. In particular, the three sets in the lemma each have twenty elements.

Formula (4.1) attaches the factor $z - z_c$ to the coset c. A small positively oriented loop around an elliptic point acts on the Hecke correspondence by permuting its cosets. This action is induced by a generator of the corresponding elliptic stabilizer, so analytic continuation carries the factor $z - z_c$ to the factor indexed by the image of c. Under the chosen splitting and the inversion convention above, reduction modulo 19 turns this coset action into the projective action stated in the lemma. □

To use the lemma, we need the splitting explicitly. Using Hensel's lemma, we get a $b \in \mathbb{Z}_{19}$ with $b^2 = -13$ and $b \equiv 5 \pmod{19}$. Take

$$I \longmapsto \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad J \longmapsto \begin{pmatrix} 4 & b \\ b & -4 \end{pmatrix}$$

in $M_2(\mathbb{Z}_{19})$. These matrices satisfy the quaternion relations given at the beginning of the section. Modulo 19 the second matrix is $\left(\begin{smallmatrix} 4 & 5 \\ 5 & -4 \end{smallmatrix}\right)$, and the images of the four displayed generators of $\mathcal{O}$ span $M_2(\mathbb{F}_{19})$. Using Nakayama's lemma, we see that the induced map $\mathcal{O} \otimes \mathbb{Z}_{19} \to M_2(\mathbb{Z}_{19})$ is bijective. Let ζ_2, ζ_4, and ζ_6 denote generators of the elliptic stabilizers of orders 2, 4, and 6 described in [Yan13, §4]. Substituting the displayed matrices for I and J gives, up to scalars,

$$\zeta_2 = I(3 + J), \qquad \zeta_4^{-1} = 1 - I, \qquad \zeta_6 = (1 - I)(3 + J).$$

The inverse at order 4 gives the required product-one relation, since

$$\zeta_2 \zeta_4^{-1} \zeta_6 = 12,$$

and the scalar 12 is trivial in PGL_2. Reducing these elements modulo 19 gives permutations of $\mathbb{P}^1(\mathbb{F}_{19})$ with cycle types

$$2^{10}, \qquad 4^5, \qquad 1^2 6^3. \tag{4.2}$$

The group these three permutations generate has order $19(19^2-1) = 6840$.

Proposition 4.2. *The geometric monodromy group of* $P(T,z)$*, in its natural degree* 20 *action, is* $\mathrm{PGL}_2(\mathbb{F}_{19})$.

Proof. By Lemma 4.1 the monodromy of the three orbifold generators is the explicit matrix action just described. Those three permutations generate a group of order $6840 = |\mathrm{PGL}_2(\mathbb{F}_{19})|$ contained in the image of $\mathrm{PGL}_2(\mathbb{F}_{19})$, so the two coincide. □

4.2. **The sign pullback and a harmonic model.** The determinants of the three elliptic stabilizers have square classes $(+,-,-)$, which are also the signs of the three permutations in (4.2). Since the matrices generate $\mathrm{PGL}_2(\mathbb{F}_{19})$, the kernel of the sign character on the monodromy is exactly $\mathrm{PSL}_2(\mathbb{F}_{19})$. The stored equation of $P(T,z)$ satisfies

$$\operatorname{disc}_z P(T,z) = (-95T)\,\Delta_0(T)^2, \qquad \Delta_0 \in \mathbb{Z}[T], \quad \deg \Delta_0 = 106,$$

for some Δ_0. Since adjoining the square root of a discriminant kills the sign character, the base change

$$T = -95s^2 \tag{4.3}$$

gives the sign double cover, and the normalization of the pullback has geometric monodromy $\mathrm{PSL}_2(\mathbb{F}_{19})$ which is connected since $\mathrm{PSL}_2(\mathbb{F}_{19})$ acts transitively on $\mathbb{P}^1(\mathbb{F}_{19})$. These arguments prove the following:

Lemma 4.3. *Let* $C \to \mathbb{P}^1_s$ *be the normalization of the pullback of* $P(T,z)$ *along* $T = -95s^2$*. Its geometric monodromy group is* $\mathrm{PSL}_2(\mathbb{F}_{19})$ *in the natural action on* $\mathbb{P}^1(\mathbb{F}_{19})$.

For the cover $C \to \mathbb{P}^1_s$ of Lemma 4.3, the branch values $T = 0, -540, \infty$ pull back under $T = -95s^2$ to

$$s = 0, \qquad s = \pm\sqrt{108/19}, \qquad s = \infty.$$

The equation P, reconstructed from Yang's algorithm, is too large to use directly in the deformation calculation (by "deform" we mean varying the point on the Hurwitz curve to obtain a new cover preserving the cycle types), so we introduce a smaller exact model and identify its monodromy below. In a suitable ordering, the cross-ratio of the branch values of this pullback is -1. We call such a configuration *harmonic*. The replacement model has as its source the elliptic curve

$$E: \quad v^2 = 9747u^3 - 18945u^2 + 11385u - 2187 \tag{4.4}$$

and covering map

$$\beta = 95 \cdot 19\rho^2 u\,\frac{g^6}{h^4} \;:\; E \longrightarrow \mathbb{P}^1_T, \qquad \rho = \frac{11250000}{116490258898219}, \tag{4.5}$$

Here $g, h \in \mathbb{Q}(E)$. Their exact coefficients are included in the supplementary computational files.

We have found this replacement model as follows. The deck transformation of the quadratic base change induces the involution

$$\iota\colon C \longrightarrow C, \qquad (s,z) \longmapsto (-s,z)$$

on the normalized sign pullback. Over finite fields, the quotient by ι has genus one, and its reductions identify isogeny class 114a as a candidate for E. The required ramification partitions translate into divisor equations for the covering function on this elliptic curve. We first find a solution to these equations over $\mathbb{F}_7$. We then lift this solution using Hensel's lemma and recover rational coefficients. Direct substitution verifies that these coefficients satisfy the defining identity. The calculation over $\mathbb{F}_7$ is only how we found the model. The proof uses the exact identity and the comparison below.

The ramification partitions above the three branch values of β are

$$\beta^{-1}(0) : 1^2 6^3, \qquad \beta^{-1}(-540) : 2^{10}, \qquad \beta^{-1}(\infty) : 4^5.$$

Three matching fiber types do not make our replacement cover (4.5) the same cover as Yang's nor determine its monodromy. By comparing the two reductions modulo 23, the next lemma proves that the harmonic model nevertheless has Yang's geometric monodromy group.

Lemma 4.4. *Let $\pi_Y\colon X_Y \to \mathbb{P}^1_T$ be Yang's degree-20 cover obtained by normalizing $P(T,z)=0$. The covers π_Y and $\beta\colon E \to \mathbb{P}^1_T$ have good tame reduction at 23. Their special fibers are isomorphic as degree 20 covers of $\mathbb{P}^1_{\mathbb{F}_{23}}$. Consequently, the harmonic cover β has geometric monodromy $\mathrm{PGL}_2(\mathbb{F}_{19})$ in its natural degree 20 action.*

Proof. Let $X_Y \to \mathbb{P}^1_T$ denote Yang's degree-20 cover, obtained by normalizing $P(T,z)=0$, and let $\beta\colon E \to \mathbb{P}^1_T$ denote the harmonic cover given above. We compare these two covers modulo 23. Put $k=\mathbb{F}_{23}$. The cubic in (4.4) remains squarefree over k and the map β retains degree 20. The displayed ramification partitions and the three branch values all remain distinct. Since its ramification indices are 2, 4, and 6, all prime to 23, the harmonic cover has good tame reduction.

The reduction $\overline{P}(T,z)$ is irreducible of degree 20 over $k(T)$. Substituting $T=\beta$ and taking the norm from $k(E)$ to $k(u)$ produces factors of degrees 2 and 38, and the greatest common divisor of the degree-two factor with $\overline{P}(\beta,z)$ is linear over $k(E)$. This exhibits an embedding $k(T)[z]/(\overline{P}) \hookrightarrow k(E)$ of $k(T)$-algebras. Both $k(T)$ algebras have degree 20 over $k(T)$, so the embedding gives an isomorphism of covers. Thus the normalized reduction of Yang's cover is the same smooth tame cover as the reduction of the harmonic model.

Now compare characteristic 0 with characteristic 23. A degree-20 monodromy group has order dividing 20!, and $23 \nmid 20!$, so the monodromy is prime to 23 and is therefore unchanged by specialization [Gro71, Exposé XIII, Cor. 2.9 and §2.10]. Both characteristic zero covers thus have the monodromy of the common special fiber, which is $\mathrm{PGL}_2(\mathbb{F}_{19})$ by Proposition 4.2. □

We have seen that Yang's cover and β have the same monodromy, but we also need to express the quadratic base change on E to eliminate the sign character. Form the fiber product

$$E \times_{\mathbb{P}^1_T} \mathbb{P}^1_s,$$

using $\beta\colon E \to \mathbb{P}^1_T$ and $T=-95s^2$, and let C_h be its normalization. The induced map $\pi_h\colon C_h \to \mathbb{P}^1_s$ is the harmonic sign pullback of the cover β. It has degree 20, and its function field is

$$\mathbb{Q}(C_h)=\mathbb{Q}(E)(s), \qquad s^2=-\beta/95.$$

Equation (4.5) gives the identity

$$-\frac{\beta}{95}=\left(\rho\frac{g^3}{h^2}\right)^2(-19u).$$

Consequently the element

$$W=\frac{sh^2}{\rho g^3}\in\mathbb{Q}(C_h)$$

satisfies

$$W^2=-19u, \qquad sh^2=\rho W g^3.$$

Conversely, the second identity recovers s from W, so we have an equality of fields $\mathbb{Q}(C_h)=\mathbb{Q}(E)(W)$. As a result, W describes the specified sign pullback, rather than a different quadratic extension of $\mathbb{Q}(E)$.

We next eliminate the elliptic coordinate v. Write $E\colon v^2=f_3(u)$ and decompose the two explicit functions g and h of 4.5 as

$$h^2=h_0(u)+vh_1(u), \qquad g^3=g_0(u)+vg_1(u),$$

with $h_i, g_i \in \mathbb{Q}(u)$. After substituting $u=-W^2/19$, the relation $sh^2-\rho W g^3=0$ becomes

$$\Phi_0(s,W)+v\Phi_1(s,W)=0,$$

where

$$\Phi_0(s,W)=sh_0(-W^2/19)-\rho W g_0(-W^2/19),$$
$$\Phi_1(s,W)=sh_1(-W^2/19)-\rho W g_1(-W^2/19).$$

Both functions Φ_0 and Φ_1 are explicit and linear in s. We multiply them by a common denominator to make them polynomials in s and W. The involution of the quadratic extension $\mathbb{Q}(W,\nu)/\mathbb{Q}(W)$ sends ν to $-\nu$. Its norm eliminates ν:

$$\mathrm{Norm}(\Phi_0+\nu\Phi_1)=(\Phi_0+\nu\Phi_1)(\Phi_0-\nu\Phi_1)=\Phi_0^2-f_3(-W^2/19)\Phi_1^2.$$

On the dense open set where $\Phi_1\neq 0$, the equation $G(s,W)=0$ recovers $\nu=-\frac{\Phi_0(s,W)}{\Phi_1(s,W)}$. The norm identity then gives $\nu^2=f_3(-W^2/19)$, so the recovered νsatisfies the original relation. Hence the primitive factor $G(s,W)$ and C_h have the same function field. After substitution, we clear the denominators of g and h. These denominators vanish at $u=1$, equivalently at $W^2+19=0$, and clearing them introduces the factor $(W^2+19)^2$. This factor is supported entirely on the denominator locus and is therefore absent from the original rational equation. Canceling it preserves the function field of the pullback. Dividing the remaining polynomial by its content and fixing its overall sign then gives the primitive integral equation

$$G(s,W)=G_2(W)s^2+G_1(W)s+G_0(W)=0. \tag{4.6}$$

It has degree 2 in s because the norm multiplies two expressions linear in s, and the exact calculation gives degree 20 in W. The coefficients of each G_i can be found in the accompanying files. Accordingly, projection to the W-line is a double cover, while projection to the s-line is the required degree-20 map π_h.

The discriminant of (4.6) as a quadratic in s is

$$\Delta_G(W)=G_1(W)^2-4G_0(W)G_2(W)=H_6(W)R_{17}(W)^2,$$

where H_6 is squarefree of degree 6 and R_{17} has degree 17. Completing the square in s and setting

$$V=\frac{2G_2(W)s+G_1(W)}{R_{17}(W)}$$

identifies the function field of the curve $G(s,W)=0$ with the function field of

$$V^2=H_6(W).$$

Thus we can remove R_{17}^2 in a birational change of coordinates. The zeros contributed by R_{17}^2 do not give branch points after normalization. The six simple roots of H_6 are the branch points of the double cover of the W-line and since $\deg H_6=6$ is even, we see that infinity contributes no further branch point. Finally, Riemann–Hurwitz formula gives

$$2g(C_h)-2=2(-2)+6=2,$$

so C_h has genus 2. Since the base change is the sign pullback, the geometric monodromy of π_h is $\mathrm{PSL}_2(\mathbb{F}_{19})$. The accompanying computation reconstructs $G(s,W)$ exactly from g and h and verifies the resulting identity coefficient by coefficient.

4.3. **A rational genus-two cover.** The harmonic cover is defined over $\mathbb{Q}$ with a special branch configuration, but specializing it does not give a totally real polynomial. We therefore move along the family of covers with the same local data until we reach a point where it does, keeping track of rationality.

Consider the Nielsen class $(2A,2A,2A,3A)$ in $\mathrm{PSL}_2(\mathbb{F}_{19})$, and let $\mathcal{H}^{\mathrm{in}}$ and $\mathcal{H}^{\mathrm{abs}}$ denote its inner and absolute Hurwitz curves. Riemann–Hurwitz applied to the passport of the exact degree-63 Belyi map shows that $\mathcal{H}^{\mathrm{abs}}$ has genus zero. We normalize a coordinate v on this curve by sending its unique cusps of widths 1, 3, and 2 to 0, 1, and ∞, respectively.

Since $N_{S_{20}}(\mathrm{PSL}_2(\mathbb{F}_{19}))/\mathrm{PSL}_2(\mathbb{F}_{19})=\mathrm{PGL}_2(\mathbb{F}_{19})/\mathrm{PSL}_2(\mathbb{F}_{19})$ has order 2, the map $\mathcal{H}^{\mathrm{in}}\to\mathcal{H}^{\mathrm{abs}}$ has degree 2. The reduced braid action shows that this map is branched precisely above $v=0$ and $v=4$. Consequently, its function field has the form

$$\mathbb{Q}(\mathcal{H}^{\mathrm{in}})=\mathbb{Q}(v)\big(\sqrt{d\,v(v-4)}\big),\qquad d\in\mathbb{Q}^\times/\mathbb{Q}^{\times 2}.$$

The fiber above $v=1$ consists geometrically of two inner cusps, which together form a degree-2 closed point over $\mathbb{Q}$. At either cusp, the degree-20 cover degenerates into two irreducible components. The induced three-point covers on these components have monodromy groups A_4 and S_3, respectively, and the components meet at a node with inertia group C_3. Complex conjugation exchanges the two cusps.

The branch cycle lemma [BF02, §3.2.2] therefore identifies the residue field of this degree-2 closed point with $\mathbb{Q}(\zeta_3) = \mathbb{Q}(\sqrt{-3})$. On the other hand, substituting $v = 1$ gives the residue field $\mathbb{Q}(\sqrt{-3d})$. Hence d is a square. After rescaling the second coordinate to absorb d, the inner Hurwitz curve has the affine model

$$\omega^2 = v(v-4).$$

This conic admits the rational parametrization

$$v = 2 + t + t^{-1}, \qquad \omega = t - t^{-1}. \tag{4.7}$$

The harmonic point is given by $t = 1$. We move to the point

$$t_* = -\frac{45}{13}, \qquad (v_*, \omega_*) = \Big(-\frac{1024}{585}, -\frac{1856}{585}\Big), \tag{4.8}$$

chosen for two reasons. First, it lies in an interval of the real inner line over which all four branch points stay real and complex conjugation acts trivially on the Nielsen class. Hence the covers above t_* can have totally real fibers. Second, it is congruent to the harmonic point modulo 29, which is what makes the Hensel lift below possible.

We first express the three finite branch values of the cover in terms of the coordinate t on the inner Hurwitz curve. Let

$$j_H \colon \mathbb{P}^1_v \longrightarrow \mathbb{P}^1_j$$

be the exact degree-63 Hurwitz Belyi map. Choose coprime polynomials $p_3, p_c \in \mathbb{Q}[v]$ such that

$$j_H(v) = \frac{256}{729}\frac{p_3(v)}{p_c(v)}.$$

Thus the zeros of p_3 lie above $j = 0$, while the zeros of p_c are the finite poles of j_H. Define $p_2 \in \mathbb{Q}[v]$ by

$$j_H(v) - 1728 = \frac{256}{729}\frac{p_2(v)}{p_c(v)}.$$

Its zeros therefore lie above $j = 1728$. These two expressions are equivalent to the exact Belyi identity

$$4p_3 - 4p_2 = 19683 p_c. \tag{4.9}$$

With this normalization, the relevant numerator factorizations are

$$p_3 = A_{20}^3 B_3, \qquad p_2 = C_{31}^2 (v-4),$$

where the subscripts record the degrees of the factors. Recall that the inner Hurwitz line has equation $\omega^2 = v(v-4)$. Define

$$a(v) = -3vA_{20}(v)B_3(v), \qquad b(v,\omega) = 2vB_3(v)C_{31}(v)\omega.$$

A direct calculation using 4.9 gives

$$\frac{4a(v)^3}{4a(v)^3 + 27b(v,\omega)^2} = \frac{j_H(v)}{1728}.$$

For a separable cubic $Y^3 + aY + b$, the double cover $\nu^2 = Y^3 + aY + b$ is branched at the three roots and ∞ and has j-invariant $1728 \cdot 4a^3/(4a^3 + 27b^2)$. Conversely, this j-invariant determines the projective equivalence class of the unordered four-point branch configuration. Therefore the equality above shows that the roots of

$$Y^3 + a(v)Y + b(v,\omega)$$

together with ∞ represent the branch configuration parametrized by the corresponding point of the Hurwitz curve.

Using the parametrization 4.7 we put

$$\mathcal{A}(t) = t^{24} a(v(t)), \qquad \mathcal{B}(t) = t^{36} b(v(t), \omega(t)). \tag{4.10}$$

The powers of t clear the denominators introduced by the parametrization. They are compatible with the change of target coordinate $Y \mapsto t^{12}Y$, and therefore do not change the associated four-point

configuration. We observe by substitution that $\mathcal{A}, \mathcal{B}$ are in $\mathbb{Q}[t]$. Thus the three finite branch values at the inner point t are the roots of

$$Y^3 + \mathcal{A}(t)Y + \mathcal{B}(t).$$

For the chosen target point $t_* = -45/13$, write

$$(A_*, B_*) = (\mathcal{A}(t_*), \mathcal{B}(t_*)).$$

We now describe the deformation chart. For a general pair (A, B), let y_1, y_2, y_3 be the roots of Y^3+AY+B. We seek a degree 20 map to the Y-line whose fibers above y_1, y_2, y_3 have cycle type 2^{10} and whose fiber above infinity has cycle type $1^2 3^6$. We represent such a map by

$$F(Y, z) = N_0(z) + YN_1(z) + Y^2N_2(z) = 0, \tag{4.11}$$

where projection to the coordinate Y is the intended covering map.

Introduce the cubic algebra

$$\mathcal{C}_{A,B} = \mathbb{Q}[\theta]/(\theta^3 + A\theta + B).$$

The element θ simultaneously represents the three finite branch values. Choose an element $\kappa \in \mathcal{C}_{A,B}$ and a monic degree 10 polynomial $U \in \mathcal{C}_{A,B}[z]$. We now define N_0, N_1, N_2 by the identity

$$\kappa\, U(\theta, z)^2 = N_0(z) + \theta N_1(z) + \theta^2 N_2(z) \qquad \text{in } \mathcal{C}_{A,B}[z]. \tag{4.12}$$

Indeed, under the embedding that sends θ to y_i, this identity becomes

$$F(y_i, z) = \kappa_i U_i(z)^2.$$

Consequently, when U_i is squarefree, the fiber above y_i consists of ten points of ramification index 2. This gives cycle type 2^{10} at each of the three finite branch values.

The fiber above $Y = \infty$ is controlled by the leading coefficient $N_2(z)$. We impose

$$N_2 = c_1 Q_6^3 Q_2, \tag{4.13}$$

where Q_6 and Q_2 are monic of degrees 6 and 2. Thus this fiber contains six points of ramification index 3 and two unramified points, which is precisely the cycle type $1^2 3^6$.

Finally, since (4.11) is quadratic in Y, its discriminant with respect to Y is

$$D(z) = N_1(z)^2 - 4N_0(z)N_2(z).$$

We impose (in order to enforce the normalization having genus 2)

$$D = c_2 H_6 R_{17}^2, \tag{4.14}$$

where H_6 and R_{17} are monic of degrees 6 and 17, respectively, and H_6 is required to be squarefree. Completing the square in (4.11) gives

$$\left(2N_2Y + N_1\right)^2 = D.$$

After dividing by R_{17}^2, the function field of the normalization is generated by

$$v^2 = c_2 H_6(z).$$

Since H_6 is a squarefree sextic, this is a smooth curve of genus 2. The factor R_{17}^2 has even multiplicity and introduces no additional branch points on the normalization.

There are 66 coordinates in this chart. The element κ and the monic polynomial U contribute $3 + 30$ coordinates. The polynomials Q_6, Q_2 and the scalar c_1 contribute $6 + 2 + 1$. Finally, c_2, H_6, R_{17} contribute $1 + 6 + 17$. The polynomials N_0, N_1, N_2 are determined by (4.12) and are not additional coordinates. Comparing coefficients in (4.13) gives 21 equations, while comparing coefficients in (4.14) gives 41. Hence the chart is defined by 62 equations in 66 coordinates.

We work on the open subchart on which $Y^3 + AY + B$ is separable, N_1 and N_2 are coprime, κ is a unit, $c_1 c_2 \neq 0$, each U_i is squarefree, H_6 is squarefree, and Q_6 and Q_2 are squarefree and coprime. On this subchart the three finite fibers have type 2^{10} and the fiber above infinity has type $1^2 3^6$. The normalization is the smooth genus-two curve $v^2 = c_2 H_6(z)$. Exact calculation verifies that both the harmonic point and the target point lie in this open subchart.

We next place the harmonic cover in this chart. The harmonic cover has branch values

$$s = 0, \qquad s = \pm\sqrt{\frac{108}{19}}, \qquad s = \infty.$$

Under the change of coordinate $Y = 2222316/s$, these become

$$Y = \infty, \qquad Y = \pm 2222316\sqrt{\frac{19}{108}}, \qquad Y = 0.$$

Put $A_0 = -868843330308$ and $B_0 = 0$. Since we have $-A_0 = 2222316^2\,\frac{19}{108}$, the three finite values are precisely the roots of $Y^3 + A_0Y + B_0$.

After making the substitutions $Y = 2222316/s$ and $z = W$, the harmonic plane model supplies explicit values of all 66 chart coordinates. Exact substitution verifies the equations (4.12), (4.13), and (4.14). Thus the harmonic cover gives an exact rational point of the deformation chart.

Modulo 29, the Jacobian of the 62 defining equations at this point has rank 62. The chart is therefore smooth of dimension 4 there. We choose four coordinate functions whose differentials complement the Jacobian of the 62 defining equations and fix their values. Adding these four conditions produces a square system of 66 equations whose Jacobian is invertible modulo 29.

Since $t_* = -\frac{45}{13} \equiv 1 \pmod{29}$, the target branch pair (A_*, B_*) reduces modulo 29 to the harmonic branch pair (A_0, B_0). Multivariate Hensel lifting therefore gives a unique solution over $\mathbb{Q}_{29}$ with branch pair (A_*, B_*), within the formal slice defined by the four chosen coordinates and in the modulo-29 residue class of the harmonic point.

It remains to prove that this 29-adic point we just constructed descends to $\mathbb{Q}$. We first reconstruct its absolute genus two invariants as rational numbers and verify the reconstructed relations exactly. Mestre's construction [Mes91] associates to these invariants a conic over $\mathbb{Q}$. In the present case this conic has an explicit rational point, which removes the obstruction to a genus-two model over $\mathbb{Q}$. Mestre's construction therefore produces a sextic model of the source curve over $\mathbb{Q}$.

The rational sextic and the sextic obtained from the point produced by Hensel lifting represent the same genus-two curve, but use different coordinates on the hyperelliptic line. Modulo 29, we enumerate all projective linear changes of coordinates that could identify their reductions. The change that identifies the two reductions is then lifted by a nonsingular four-variable Hensel Lemma calculation. The three variables describe the projective linear transformation, and the fourth records the proportionality factor between the two sextics. These computations can be found in the accompanying files.

We then verify that the lifted transformation carries the marked divisors of degrees 2 and 6 above infinity to the corresponding divisors on the rational model. Rational reconstruction gives the 63 coefficients of N_0, N_1, N_2. Substitution over $\mathbb{Q}$ confirms the branch pair (A_*, B_*), the four fixed coordinates, and the chosen reduction modulo 29. Hensel uniqueness under these conditions identifies this rational point with the lift obtained from the harmonic cover.

Let $(N_{0,*}, N_{1,*}, N_{2,*})$ denote the resulting rational coefficient rows. These rows and the corresponding pair (A_*, B_*) are included in the accompanying files. Put

$$F_*(Y, z) = N_{0,*}(z) + YN_{1,*}(z) + Y^2N_{2,*}(z).$$

This fixes a single cover. The stored rational point has branch pair (A_*, B_*), satisfies all 62 identities and the same four fixed coordinates exactly, and reduces modulo 29 to the selected point, including the marked divisors Q_2 and Q_6. It is therefore the Hensel lift in the chosen residue class and formal slice.

For F_*, the branch cubic is irreducible and squarefree, the three involution fibers are exact squares of degree 10, N_2 factors as $Q_6^3Q_2$, and H_6 is squarefree. The visible ramification is

$$3 \cdot 10 + 6(3 - 1) = 42 = 2 \cdot 20 + 2 \cdot 2 - 2,$$

and the source cuve has genus 2 so Riemann–Hurwitz shows that these four fibers are the whole branch locus.

Proposition 4.5. *The cover $F_*(Y, z) = 0$ has smooth genus-two source and geometric monodromy $\mathrm{PSL}_2(\mathbb{F}_{19})$.*

Proof. The identities above along with the Riemann–Hurwitz formula prove smoothness, the genus, and completeness of the branch locus. For the monodromy, note that the harmonic point and the target reduce to the same nonsingular point of the chart modulo 29. Away from the four disjoint branch sections, the degree 20 family is finite étale. Any monodromy group here is a subgroup of S_{20}, so its order divides 20! and is prime to 29. Therefore specialization preserves the monodromy [Gro71, Exposé XIII, Cor. 2.9 and §2.10] and both characteristic zero fibers have the monodromy of the common special fiber. By Lemmas 4.3 and 4.4, and the identification of W made at the end of §4.2, that group is $\mathrm{PSL}_2(\mathbb{F}_{19})$. □

Remark 4.6. In the language of Appendix A, the covers in this chart lie in the Nielsen class of $\mathrm{PSL}_2(\mathbb{F}_{19})$ with three branch cycles of order 2, of cycle type 2^{10}, and one of order 3, of cycle type $1^2 3^6$. The analog of this class at $\ell = 13$ is $(2A, 2A, 2A, 3A)$, and that is precisely the class König rules out [Kön17, §7]. In that case, the source curves have genus 0, however the Hurwitz curve is an elliptic curve of rank 0 whose rational points give no covers with real fibers. At $\ell = 13$, we therefore moved to $(2A, 2A, 2A, 6A)$ and genus one sources. At $\ell = 19$, we use the class $(2A, 2A, 2A, 3A)$, in which the Hurwitz curve has genus 0 and the source curves have genus 2.

4.4. **The Galois group of f_{19}.** We specialize F_* at

$$Y_0 = -7546608500347.$$

We then apply a change of the primitive element for the specialization and follow it by polynomial reduction. The result is the monic polynomial f_{19} of Appendix B. Using a minimal polynomial identity, we have verified that f_{19} and $F_*(Y_0, z)$ define isomorphic degree 20 fields.

We chose the point Y_0 as follows. The three roots of $Y^3 + A_*Y + B_*$ divide the real Y-line into four intervals. An exact Sturm calculation [BPR06, §2.2.2] shows that only one of these intervals gives specializations with twenty real roots. We searched this interval subject to congruence conditions at 3 and 17 chosen to produce favorable local data for Böge's criterion. The Chinese remainder theorem reduces the search to an arithmetic progression which contains Y_0.

Proposition 4.7. *The polynomial f_{19} is irreducible, all twenty of its roots are real, and*

$$\mathrm{Gal}(L_{19}/\mathbb{Q}) \cong \mathrm{PSL}_2(\mathbb{F}_{19})$$

in the natural action on $\mathbb{P}^1(\mathbb{F}_{19})$.

Proof. Exact factorization over $\mathbb{F}_7$ and over $\mathbb{F}_{23}$ gives

$$\begin{aligned} f_{19} \bmod 7 &= (x^5 + x^3 + 2x^2 + 4x + 2)(x^5 + 5x^4 + 5x^3 + 5x + 4) \\ &\quad \cdot (x^5 + 6x^4 + 2x^3 + 5x^2 + 4x + 6)(x^5 + 6x^4 + 4x^3 + 3x^2 + 5x + 6), \\ f_{19} \bmod 23 &= (x + 17)(x^{19} + 2x^{18} + 12x^{17} + 4x^{16} + 16x^{15} + 6x^{13} + x^{11} \\ &\quad + 20x^9 + x^8 + 14x^7 + 7x^6 + 16x^5 + 9x^4 + 21x^3 + 7x^2 + 5x + 22), \end{aligned}$$

both squarefree with every displayed factor irreducible. The degrees of the factors of a rational factor of f_{19} would have to add up to the same number in both patterns. The subset sums of $\{5, 5, 5, 5\}$ are $0, 5, 10, 15, 20$ and those of $\{1, 19\}$ are $0, 1, 19, 20$, whose only common values are 0 and 20. Hence f_{19} is irreducible. By Dedekind's theorem its Galois group G contains an element of cycle type $1 \cdot 19$, hence an element of order 19.

Irreducibility shows that G is transitive. Let $H = \mathrm{PSL}_2(\mathbb{F}_{19})$ in its natural degree-20 action, and let A be the arithmetic monodromy group of F_*. Proposition 4.5 gives the geometric monodromy group H. Hence

$$H \trianglelefteq A \leqslant N_{S_{20}}(H).$$

The exact normalizer calculation gives

$$N_{S_{20}}(H) = \mathrm{PGL}_2(\mathbb{F}_{19}), \qquad N_{S_{20}}(H) \cap A_{20} = H.$$

Since Y_0 is not a branch point of the family, specialization at Y_0 embeds G into the arithmetic monodromy group [Ser08, Ch. 1]. We also check that the discriminant of f_{19} is a nonzero square, so $G \leqslant A_{20}$ and therefore $G \leqslant H$.

For the lower bound, an exhaustive search inside H shows that no proper subgroup of H containing an element of order 19 is transitive. We conclude $G = H$.

Additionally, the exact Magma script in the level 19 code package proves irreducibility and, using `GaloisProof`, identifies the Galois group in its degree-20 action with $\mathrm{PSL}_2(\mathbb{F}_{19})$. Sturm arithmetic in the accompanying Sage file gives twenty real roots [BPR06, §2.2.2]. □

4.5. **Ramification.** Write $K_{19} = \mathbb{Q}[x]/(f_{19})$, put $p_3 = 5605357159$, and let

$$p_4 = 136462247050703576496865693263674826888464372857556592926617399 0\\ 856504046452283837355063452317992591,$$

a prime of 100 digits. A certified maximal-order computation gives

$$\mathrm{disc}(K_{19}) = 17^{10}\, p_3^{16}\, p_4^{10}. \tag{4.15}$$

Lemma 4.8. *The primes ramified in L_{19} are exactly 17, p_3 and p_4, and for each the pair $(e(q), f(q))$ is as in Table 2.*

q	type of q in K_{19}	(I, D)	$(e(q), f(q))$
17	$(2,5)^2$	(C_2, C_{10})	$(2,5)$
p_3	$(1,1)^2, (9,1)^2$	(C_9, C_9)	$(9,1)$
p_4	$(2,2)^5$	$(C_2, C_2 \times C_2)$	$(2,2)$

TABLE 2. Local data for K_{19} and for its Galois closure L_{19}.

Proof. As in Lemma 3.2, we construct the maximal order of K_{19} using PARI, and certify it with `nfcertify`, and also confirm the discriminant (4.15). Exact factorization of the primes $17\mathcal{O}_{K_{19}}$, $p_3\mathcal{O}_{K_{19}}$ and $p_4\mathcal{O}_{K_{19}}$ gives the decomposition types recorded in the second column of Table 2. All three ramified primes are tame, so the enumeration of pairs $I \trianglelefteq D \leqslant \mathrm{PSL}_2(\mathbb{F}_{19})$ described after Lemma 2.2 applies with I cyclic, D/I cyclic. In each case, we find a single conjugacy class in the stated permutation action. Every prime ramified in K_{19} ramifies in L_{19}, and by Lemma 2.3 there is no further ramification, proving the assertion. □

The prime p_3 has $e(p_3) = 9$ odd and so imposes no condition in Theorem 2.1. The other two give

$$e(17) = 2, \quad f(17) = 5 \text{ odd}, \quad 17 \equiv 1 \pmod 4,$$

and

$$e(p_4) = 2, \quad f(p_4) = 2 \text{ even}, \quad p_4 \equiv 3 \pmod 4.$$

Both satisfy the parity condition, so Lemma 4.8 and Proposition 4.7 together prove Theorem 1.2 for $\ell = 19$. This completes the proof of Theorem 1.1. □

5. A tower of totally real congruence extensions

The representations produced by Theorem 1.1 are even representations which are known to be more difficult to study. In two papers, [Ram99, Ram02], Ramakrishna investigated obstructions to lifting mod p Galois representations to characteristic 0 p-adic representations. His deformation theorem, combined with Theorem 1.1, gives the following consequence.

Corollary 5.1. *For each $\ell \in \{13, 19\}$ there is a continuous surjection*

$$\rho_\ell \colon \mathrm{Gal}_{\mathbb{Q}} \twoheadrightarrow \mathrm{SL}_2(\mathbb{Z}_\ell)$$

unramified outside a finite set of primes, whose reduction modulo ℓ cuts out M_ℓ, for which every complex conjugation lies in $\ker \rho_\ell$. Consequently, the fixed field $M_{\ell,n}$ of $\ker(\rho_\ell \bmod \ell^n)$ is totally real with

$\mathrm{Gal}(M_{\ell,n}/\mathbb{Q}) \cong \mathrm{SL}_2(\mathbb{Z}/\ell^n\mathbb{Z})$, *the fields satisfy* $M_{\ell,n} \subset M_{\ell,n+1}$, *and all are unramified outside one fixed finite set.*

Proof. Fix an isomorphism $\mathrm{Gal}(M_\ell/\mathbb{Q}) \simeq \mathrm{SL}_2(\mathbb{F}_\ell)$ and let $\overline{\rho}_\ell\colon \mathrm{Gal}_\mathbb{Q} \twoheadrightarrow \mathrm{SL}_2(\mathbb{F}_\ell)$ be the corresponding representation. Ramakrishna proved that for a finite field k of characteristic at least 7, every surjection $\mathrm{Gal}_\mathbb{Q} \twoheadrightarrow \mathrm{SL}_2(k)$ admits a deformation to $\mathrm{SL}_2(W(k))$ that is unramified outside a finite set [Ram02, Corollary 1(a)]. No parity hypothesis on the residual representation is imposed. Since $W(\mathbb{F}_\ell) = \mathbb{Z}_\ell$, this gives ρ_ℓ.

For $\ell \geqslant 5$, a closed subgroup of $\mathrm{SL}_2(\mathbb{Z}_\ell)$ that surjects modulo ℓ is the whole group [Ser68, Ch. IV, Lemma 3], so the lift is surjective. Let c be a complex conjugation. The total reality of M_ℓ gives $\overline{\rho}_\ell(c) = I$, so $\rho_\ell(c) \equiv I$ modulo ℓ and $\rho_\ell(c) + I$ is invertible. Since $c^2 = 1$, we also have $\rho_\ell(c)^2 = I$. Therefore

$$\big(\rho_\ell(c) - I\big)\big(\rho_\ell(c) + I\big) = 0,$$

which forces $\rho_\ell(c) = I$. Reducing modulo ℓ^n gives the stated finite quotients, all totally real and all unramified outside the ramification set of ρ_ℓ. The kernels decrease with n, so their fixed fields are nested. □

Remark 5.2. Corollary 5.1 is a pure existence result. It produces no defining polynomials for the higher layers, and we do not assert that Ramakrishna's even lifts are potentially semistable or geometric.

6. The computations

All the necessary code can be found in the following GitHub repository.

https://github.com/eekarabiyik/SL2F13.

This work includes a large amount of computer assisted calculations. These consist of factorizations of integers and of polynomials over $\mathbb{Q}$ and over finite fields, Sturm counts [BPR06, §2.2.2], maximal orders and their prime factorization data, resolvent constructions, exhaustive searches in finite permutation groups, and substitutions into polynomial identities. We used SageMath 10.9 [The26b] with PARI/GP 2.17.4 [The25] for the arithmetic, and GAP 4.16.0 [The26a] with the transitive groups library [Hul23] for the group theory. Every maximal order is certified by PARI's `nfcertify`. None of the discriminants or prime factorizations used in this paper is conditional on the generalized Riemann hypothesis. Two Magma [BCP97] scripts verify Propositions 3.1 and 4.7.

To find the equations of Appendix A and §4.3, our computations involved numerical continuation, p-adic lifting, Berlekamp–Massey and Padé reconstruction, and rational reconstruction. The resulting candidates were retained only after exact substitution into their defining identities.

The archive in the GitHub repository contains one self-contained package for each prime. Each has a short certificate that begins with the displayed coefficients and verifies everything asserted about L_ℓ in §3 and §4. The longer level-13 check verifies the exact degree-54 Belyi identity, the stored genus-one family, the specialization at t_0, and the relation with the displayed polynomial. It does not repeat the numerical continuation and Hensel computations by which the stored family was discovered. The longer level-19 check reconstructs Yang's equation from Hecke operators, the harmonic model, the sign pullback, and the 29-adic descent.

Appendix A. The construction of f_{13}

This appendix records how f_{13} was computed.

A.1. **The Hurwitz component.** Let $G = \mathrm{PSL}_2(\mathbb{F}_{13})$ act on fourteen points. A four-point G-cover of $\mathbb{P}^1$ is described topologically by branch cycles $g_1, g_2, g_3, g_4 \in G$ with

$$g_1 g_2 g_3 g_4 = 1, \qquad \langle g_1, g_2, g_3, g_4 \rangle = G,$$

taken up to simultaneous conjugation. The resulting finite set is the inner Nielsen class. The Hurwitz braid group acts on it by moving the branch points. The orbits of this action are the connected components of the corresponding Hurwitz space. We refer to [Kön17, §2] for further details including the algorithms used.

We take the Nielsen class

$$g_1, g_2, g_3 \in 2A, \qquad g_4 \in 6A, \tag{A.1}$$

whose two conjugacy classes act on $\mathbb{P}^1(\mathbb{F}_{13})$ with cycle types $1^2 2^6$ and $1^2 6^2$. The permutation indices sum to $3\cdot 6+10 = 28 = 2\cdot 14+2\cdot 1-2$, so by Riemann–Hurwitz the degree 14 source curves have genus 1. Note that König investigates a different obstructed genus-zero family of type $(2A, 2A, 2A, 3A)$ [Kön17, §7].

For one fixed ordering, fixing g_1 leaves 1296 tuples in (A.1). The centralizer of g_1, of order 12, acts freely. Indeed, an element fixing a generating tuple centralizes all its entries and therefore lies in the trivial center of G. Thus there are 108 inner classes. Let q_1, q_2, q_3 be the standard Hurwitz generators, where q_i interchanges the ith and $(i+1)$st branch points, and put

$$Q' = \langle q_1 q_3^{-1}, (q_1 q_2 q_3)^2 \rangle.$$

This is the standard Klein-four reduction subgroup, its image consists of the identity and the three double transpositions of the four branch labels of our Nielsen class. Its orbits identify the four markings of a branch configuration having the same cross ratio. To apply the full braid group, we use all four placements of the $6A$ class, obtaining 432 inner states. The group Q' has free orbits of size 4 so we are again left with 108 reduced inner states. The outer involution of $\mathrm{PGL}_2(\mathbb{F}_{13})/G$ identifies these in pairs, leaving 54 reduced absolute states. These states are the sheets of the map from the absolute Hurwitz curve to the branch-configuration line. Thus this map is a degree-54 Belyi map. Braid enumeration gives its three-point monodromy triple with cycle types

$$3^{18}, \qquad 1^2 2^{26}, \qquad 2\,3^4\,6\,7^3\,13. \tag{A.2}$$

The indices sum to $36+26+44 = 106 = 2\cdot 54-2$, so the compactified absolute Hurwitz curve has genus 0. Its cusps of widths 2, 6 and 13 are rational and unique. We normalize a coordinate x by placing them at 0, 8 and ∞, respectively.

The reconstruction produced the following exact degree-54 Belyi map with passport (A.2):

$$\beta = -\frac{\Phi^3}{2^6 3^9\, c}, \qquad \beta - 1 = -\frac{\Psi^2 \lambda}{2^6 3^9\, c}, \tag{A.3}$$

where $\Phi, \Psi \in \mathbb{Z}[x]$ are monic of degrees 18 and 26, given in GitHub, and

$$\lambda = x^2 - 10x + 13, \qquad \sigma = x^3 - 11x^2 + 24x - 13,$$
$$\rho = x^4 - 19x^3 + 113x^2 - 221x + 169, \quad c = x^2(x-8)^6 \rho^3 \sigma^7.$$

The two expressions in (A.3) are compatible from the equation

$$\Phi^3 - \Psi^2 \lambda = -2^6 3^9 c \tag{A.4}$$

holding in $\mathbb{Z}[x]$. Further computations show that the zero and pole divisors of β and $\beta - 1$ give the three cycle types of (A.2). Riemann–Hurwitz then shows that β has no other critical points. The map β was found as follows. We first normalized the three distinguished cusps. The resulting equations were then solved numerically to high precision and then the rational coefficients were recognized. A separate path continuation matched the monodromy triple, up to simultaneous conjugacy, with the enumerated braid triple whose cycle types are given in (A.2).

A.2. **Constructing the degree 14 family.** Under the numerical identification just described, the two simple points of β above 1, namely the roots of $\lambda(x)$, form the branch divisor of the inner-to-absolute double cover. Its function field therefore has the form

$$\mathbb{Q}(x)\big(\sqrt{d\,\lambda(x)}\big), \qquad d \in \mathbb{Q}^\times/\mathbb{Q}^{\times 2}.$$

The rational width-two cusp lies above $x = 0$. Since $\lambda(0) = 13$, the existence of a rational point above $x = 0$ implies that $13d$ is a square in $\mathbb{Q}$. Thus d has square class 13. After rescaling the second coordinate, the inner curve has the equation

$$y^2 = 13\lambda(x) = 13(x^2 - 10x + 13).$$

The rational point $(0, 13)$ gives the parametrization

$$x(\eta) = -\frac{26\eta}{\eta^2 - 10\eta + 12}, \qquad y(\eta) = \frac{13(12 - \eta^2)}{\eta^2 - 10\eta + 12}.$$

The deck involution in this parametrization becomes $\eta \mapsto 12/\eta$. We have used numerical constructions to obtain the above identities, but we also verify all the equations directly over $\mathbb{Q}$. Above the inner point η, the three order 2 branch points of the associated degree 14 cover are the roots of

$$b_\eta(T) = T^3 - 39\,\Phi(x(\eta))\,T + 26\,\Psi(x(\eta))\,y(\eta).$$

To see that this cubic represents the required point of the moduli space, apply the four-point classification used in §4.3. The double cover branched at the roots of $T^3 + pT + q$ and at ∞ has normalized j-invariant $4p^3/(4p^3 + 27q^2)$; substituting $p = -39\Phi$ and $q = 26\Psi y$ and using $y^2 = 13\lambda$ and (A.4) returns $\beta(x)$.

We work at $\eta = 1$, where $x = -26/3$ and $y = 143/3$, so that

$$b_1(T) = T^3 - \frac{3046403144842857396142588176 16}{129140163}\,T + \frac{33628775683151066628816772460184130639589 1328}{7625597484987}.$$

This cubic is irreducible with positive discriminant, its roots are real and distinct. We seek a degree-14 cover whose three finite branch values are the roots t_1, t_2, t_3 of $b_1(T)$. Above each t_i we require six points of ramification index 2 and two unramified points. The two unramified points in the prescribed fiber above $T = \infty$ form an effective divisor of degree 2 on the genus-one source. Riemann–Roch gives a nonconstant function z with this polar divisor. Hence the function field of the source is quadratic over $\mathbb{Q}(z)$, so T satisfies a quadratic equation over $\mathbb{Q}(z)$. We use this choice of z throughout the deformation. To impose the three finite factorizations simultaneously, put

$$\mathcal{C}_1 = \mathbb{Q}[\theta]/(b_1(\theta))$$

and look for a cover of the form

$$F(T, z) = N_0(z) + TN_1(z) + T^2N_2(z), \qquad (\deg N_0, \deg N_1, \deg N_2) = (14, 13, 12), \tag{A.5}$$

and require

$$F(\theta, z) = \kappa(\theta)U(\theta, z)^2V(\theta, z) \qquad \text{in } \mathcal{C}_1[z],$$

where U and V are monic of degrees 6 and 2. Under an embedding $\theta \mapsto t_i$, this becomes

$$F(t_i, z) = \kappa_i U_i(z)^2 V_i(z).$$

When U_i and V_i are squarefree and coprime, the six roots of U_i give the six ramification points of index 2, while the two roots of V_i give the two unramified points. Thus the fiber has cycle type $1^2 2^6$.

At $T = \infty$ we require two ramification points of index 6 and two unramified points. Writing $S = 1/T$ and multiplying the equation by S^2 shows that the finite part of the fiber at infinity is cut out by $N_2(z)$. We therefore impose

$$N_2(z) = Q_\infty(z)^6, \qquad \deg Q_\infty = 2. \tag{A.6}$$

With $S = 1/T$, local homogenization shows that the two distinct roots of the squarefree polynomial Q_∞ yield two branches of ramification index 6, while the degree drops of N_1 and N_2 yield two unramified branches over $z = \infty$. Thus the fiber has cycle type $1^2 6^2$.

Since (A.5) is quadratic in T, projection to the z-line makes the normalization of $F = 0$ hyperelliptic, and the genus one condition (branching in exactly four points) reads

$$N_1(z)^2 - 4N_0(z)N_2(z) = d_0\,H_4(z)\,R_{11}(z)^2 \tag{A.7}$$

with H_4 monic squarefree of degree 4 and R_{11} monic of degree 11. The desired domain is then the curve given by $Y^2 = d_0 H_4(z)$.

We work on the open subchart on which κ is a unit, $d_0 \neq 0$, each pair U_i, V_i is squarefree and coprime, and Q_∞ and H_4 are squarefree. Since b_1 is separable, these conditions give the stated fiber types and a smooth genus-one normalization.

Expanding everything in the basis $1, \theta, \theta^2$ leaves

$$\underbrace{3}_{\kappa} + \underbrace{6\cdot 3}_{U} + \underbrace{2\cdot 3}_{V} + \underbrace{2}_{Q_\infty} + \underbrace{1}_{d_0} + \underbrace{4}_{H_4} + \underbrace{11}_{R_{11}} = 45$$

unknowns. Coefficient comparison in (A.6) and (A.7) gives 44 scalar equations. At the harmonic seed their Jacobian has rank 43, because one of the 44 differential rows lies in the span of the others. The resulting two-dimensional freedom among the 45 unknowns is exactly the freedom to translate and rescale the auxiliary coordinate z. For the local continuation, we retain 43 independent equations and fix the constant and linear coefficients of Q_∞ at their seed values. The resulting 45 equations have a nonsingular Jacobian at the seed, and every exact endpoint is checked against all 44 original equations.

After continuing to $\eta = 7/2$, we make an affine change of z and impose

$$\mathrm{coef}_{Q_\infty}(z) = 0, \qquad \mathrm{coef}_{H_4}(z^3) = -4,$$

where $\mathrm{coef}_f(z^k)$ denotes the coefficient of z^k in f. The first condition fixes the translation of z, and the second fixes its scaling. This normalization is used in the subsequent reconstruction. A starting point comes from the harmonic point $\eta = 2\delta$ of the inner conic where $\delta = \sqrt{3}$. Let w be the coordinate of the codomain of the harmonic cover and X the affine coordinate in the plane model of the domain. The equation

$$457w^2X^{12}A_2(X) - (493 - 88\delta)B_2(X)C_3(X)^4 = 0$$

with

$$\begin{aligned} A_2(X) &= 481X^2 + (123\delta + 180)X + (594 - 27\delta),\\ B_2(X) &= 457X^2 - (213\delta + 648)X + (594 + 81\delta),\\ C_3(X) &= 1403X^3 - (2703\delta + 153)X^2 - (1458\delta + 1512)X - (4293\delta + 243) \end{aligned}$$

defines a cover with the required fibers. Its connectedness is checked by exact factorization, its discriminant is checked to be a square, and its normalized source has the genus-one model $Y^2 = \frac{493-88\delta}{457}A_2(X)B_2(X)$. On this normalization the two coordinates are related, after choosing the sign of Y, by

$$w = \frac{YC_3(X)^2}{X^6A_2(X)}.$$

The projection from the local solution curve to the η-line has two local branches. Numerical continuation from the harmonic solution chose one of the two branches and proposed a rational point above $\eta = 7/2$ (chosen because of proximity to $2\sqrt{3}$). We verified by exact substitution that this rational point satisfies the complete system. Starting from this point, we computed the solution in a finite field power series ring modulo 1000003 to precision 256. Berlekamp–Massey and Padé reconstruction then produced a candidate above $\eta = 1$, where the branch cubic is irreducible. We used Hensel's lemma to lift this candidate at fixed $\eta = 1$ and reconstructed all 45 coordinates over $\mathbb{Q}$. We then verified the complete system is satisfied. Hence, the finite field continuation was only used to locate the candidate at $\eta = 1$.

At the exact solution, d_0 is nonzero, H_4 is squarefree, the three pairs U_i, V_i are squarefree and coprime, every κ_i is nonzero, and Q_∞ is squarefree. Thus the three finite fibers have type $1^2 2^6$, and the fiber above infinity has type $1^2 6^2$. The visible ramification is $3 \cdot 6 + 10 = 28$, the full Riemann–Hurwitz total for a degree-14 map from a genus-one curve, so there are no other branch points. The polynomial $F(0, z)$ is irreducible, and since the leading coefficient of F in z is a nonzero constant, F is irreducible over $\mathbb{Q}(T)$.

A.3. **Specializing.** Let $\tau_1 < \tau_2 < \tau_3$ be the roots of b_1. Specializing T inside the interval (τ_2, τ_3) was numerically observed to keep all fourteen roots real. Numerically, we have

$$\tau_2 \approx 2.8040063127 \cdot 10^{10}, \qquad \tau_3 \approx 2.8043085841 \cdot 10^{10}.$$

For an integer t in that interval let $g_t(z) \in \mathbb{Z}[z]$ be the primitive part of $N_0(z) + tN_1(z) + t^2N_2(z)$, that is, the integral polynomial obtained by clearing denominators and dividing out the content. We chose

$$t_0 = 28040147150,$$

for which

$$b_1(t_0) = -\frac{2^3 \cdot 5119 \cdot p_1 \cdot p_2}{7625597484987}. \tag{A.8}$$

The three odd factors are prime and congruent to 3 modulo 4, which is what made this specialization a promising candidate. Their factorization type in the stem predicted the favorable local row $(e, f) = (2, 2)$.

Our code checked that g_{t_0} is primitive and irreducible with fourteen real roots and square discriminant. It is not monic, and its coefficients are large. Applying PARI's exact `polredabs` to $\mathbb{Q}[z]/(g_{t_0})$ produces the monic polynomial f_{13} of Example 1.3.

Appendix B. The degree 20 projective stem field

The coefficients of f_{19} are listed below.

i	a_i in $f_{19}(x) = \sum_{i=0}^{20} a_i x^i$
20	1
19	−4
18	−28733023820957366452894205043351419234
17	−14273707955962177038650090044958006232528983921598922884
16	32344407414048794495183187576247912490430844121875890141339789256519149857 7
15	30505744656740724848378140592050350959697703697714600576022983590688768852927976631 8813484256
14	−17440026067479232206936908376859538853684191246912268867416719704810975413832019982 0816069855118671399116560166 3
13	−23849624921438389957042167648259429335783437647978229458914763661615378664314329468 1841191695782618875237853323720570230642731310 2
12	43537968498731651123321745034355777967813104575455427020810791249462252686395768917 8760572368726065855319992770196934069048233657665424315823966252 5
11	83010408709381504980531227265250461515421868808923242394616754634428600871579000242 5188602151169562247586008561424425103457809287835423948961280796992258077588341412 4
10	−35262702651099038955565374901075419355338587374813496266777251752848158543306667995 9701830431393142733870692452835543389855223448709362658905393556357640534211179774612 4098936390694883
9	−12628143360269520443672033094048820308466567791132728996438726235542352558369551929 4607046317283487237765734389554120279706350426923350494613007518879362264429384071339 7889764332995692191021750836340950 6
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6	25340965260783797991006707298893603224325321460383056639630421540913003254591978224 3961440092845796642949829104967080410311261861125887661199906668833150913450927852470 3475993764481360353112568014214944807838088061952106043714992470827784436896384280357719
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3	27730867994388149287341413347498663452262152066389604634947811184353101454312687191 4303573904477889844543948010347420244208248339438837694622537839955544720182837992411 9298037716460298818843628127045806570809719952807839746192497711271967770129741224868 8785465142205229388165452989403780134722959237115210980 00
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i	a_i (continued)
1	$-151022060485038471821500791324341919090017220093854623778721541550814125135001812846$ $333069795453369161233486254043216088213440132090580316787086474103391121282087725878$ $361637509357947027223227111869926850419882752971328190012775984382618962591254350411$ $648299179628105341310932955519940794906946989016617050423688740095207071796191852083918440000$
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References

[BCP97] Wieb Bosma, John Cannon, and Catherine Playoust, *The Magma algebra system. I. the user language*, Journal of Symbolic Computation **24** (1997), no. 3–4, 235–265. ↑3, 17

[BF02] Paul Bailey and Michael D. Fried, *Hurwitz monodromy, spin separation and higher levels of a modular tower*, Arithmetic fundamental groups and noncommutative algebra, 2002, pp. 79–220. MR1935406 ↑12

[BK12] Armand Brumer and Kenneth Kramer, *Arithmetic of division fields*, Proceedings of the American Mathematical Society **140** (2012), no. 9, 2981–2995. ↑4

[Bög90] Sigrid Böge, *Witt-Invariante und ein gewisses Einbettungsproblem*, Journal für die reine und angewandte Mathematik **410** (1990), 153–159. ↑4

[BPR06] Saugata Basu, Richard Pollack, and Marie-Françoise Roy, *Algorithms in real algebraic geometry*, Second, Algorithms and Computation in Mathematics, vol. 10, Springer-Verlag, Berlin, 2006. ↑6, 15, 16, 17

[But93] Gregory Butler, *The transitive groups of degree fourteen and fifteen*, Journal of Symbolic Computation **16** (1993), no. 5, 413–422. ↑6

[CCN+85] John H. Conway, Robert T. Curtis, Simon P. Norton, Richard A. Parker, and Robert A. Wilson, *Atlas of finite groups: Maximal subgroups and ordinary characters for simple groups*, Clarendon Press, Oxford, 1985. ↑3

[Cre97] Teresa Crespo, *Galois representations, embedding problems and modular forms*, Collectanea Mathematica **48** (1997), no. 1–2, 63–83. ↑2

[Gro71] Alexander Grothendieck, *Revêtements étales et groupe fondamental (SGA 1)*, Lecture Notes in Mathematics, vol. 224, Springer-Verlag, Berlin–New York, 1971. Augmenté de deux exposés de M. Raynaud. ↑10, 15

[Hul23] Alexander Hulpke, *TransGrp: Transitive groups library*, 2023. Version 3.6.5. ↑3, 6, 17

[Klü00] Jürgen Klüners, *A polynomial with Galois group* $\mathrm{SL}_2(11)$, Journal of Symbolic Computation **30** (2000), no. 6, 733–737. ↑1, 2, 4

[Kön17] Joachim König, *Computation of Hurwitz spaces and new explicit polynomials for almost simple Galois groups*, Mathematics of Computation **86** (2017), no. 305, 1473–1498. ↑15, 17, 18

[Mes90] Jean-François Mestre, *Extensions régulières de* $\mathbb{Q}(T)$ *de groupe de Galois* $\widetilde{A}_n$, Journal of Algebra **131** (1990), no. 2, 483–495. ↑1

[Mes91] ———, *Construction de courbes de genre 2 à partir de leurs modules*, Effective methods in algebraic geometry, 1991, pp. 313–334. ↑14

[Mes94a] ———, *Annulation, par changement de variable, d'éléments de* $\mathrm{Br}_2(k(x))$ *ayant quatre pôles*, Comptes rendus de l'Académie des sciences. Série I, Mathématique **319** (1994), no. 5, 529–532. ↑3

[Mes94b] ———, *Construction d'extensions régulières de* $\mathbb{Q}(T)$ *à groupes de Galois* $\mathrm{SL}_2(\mathbb{F}_7)$ *et* $\widetilde{M}_{12}$, Comptes rendus de l'Académie des sciences. Série I, Mathématique **319** (1994), no. 8, 781–782. ↑1

[Pla07] Bernat Plans, *Generic Galois extensions for* $\mathrm{SL}_2(\mathbb{F}_5)$ *over* $\mathbb{Q}$, Mathematical Research Letters **14** (2007), no. 3, 443–452. ↑1

[Ram02] Ravi Ramakrishna, *Deforming Galois representations and the conjectures of Serre and Fontaine–Mazur*, Annals of Mathematics (2) **156** (2002), no. 1, 115–154. ↑3, 16, 17

[Ram99] ———, *Lifting Galois representations*, Inventiones Mathematicae **138** (1999), no. 3, 537–562. ↑16

[Ser08] Jean-Pierre Serre, *Topics in Galois theory*, Second, Research Notes in Mathematics, vol. 1, A K Peters, Wellesley, MA, 2008. ↑16

[Ser68] ———, *Abelian* l*-adic representations and elliptic curves*, W. A. Benjamin, New York–Amsterdam, 1968. ↑17

[Ser84] ———, *L'invariant de Witt de la forme* $\mathrm{Tr}(x^2)$, Commentarii Mathematici Helvetici **59** (1984), no. 4, 651–676. ↑4

[Son80] Jack Sonn, $\mathrm{SL}(2,5)$ *and Frobenius Galois groups over* $\mathbb{Q}$, Canadian Journal of Mathematics **32** (1980), no. 2, 281–293. ↑1

[The25] The PARI Group, *PARI/GP, version 2.17.4*, Université de Bordeaux, 2025. ↑3, 17

[The26a] The GAP Group, *GAP—groups, algorithms, and programming, version 4.16.0*, 2026. Official release 4.16.0; the certificate transcript records and checks the runtime version and the loaded TransGrp version. ↑3, 17

[The26b] The Sage Developers, *SageMath, the Sage mathematics software system, version 10.9*, 2026. Version 10.9. ↑3, 17

[Utt25] Peter Vang Uttenthal, *Level-raising of even representations of tetrahedral type and equidistribution of lines in the projective plane*, 2025. arXiv:2310.14352, version 2, 22 December 2025. ↑1

[Yan12] Yifan Yang, *Computing modular equations for Shimura curves*, 2012. arXiv:1205.5217. ↑3, 7

[Yan13] ———, *Schwarzian differential equations and Hecke eigenforms on Shimura curves*, Compositio Mathematica **149** (2013), no. 1, 1–31. ↑7, 8

[Zyw15] David Zywina, *The inverse Galois problem for* $\mathrm{PSL}_2(\mathbb{F}_p)$, Duke Mathematical Journal **164** (2015), no. 12, 2253–2292. ↑1

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